Scalability & Future Plan

Date	26 October 2026
Team ID	
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	1 Mark

Scalability & Future Plan:

This plan describes realistic ways to extend the current individual Tableau Public and static website solution. Proposed upgrades are future work, not features already implemented.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Static CSV-based data source; updates require preparing and reconnecting a refreshed file.	New data is not automatically ingested; insights may become outdated.	High
2	Tableau Public workbook and static GitHub Pages website depend on hosted services and internet access.	Limited control over hosting, access management and backend processing.	Medium
3	The supplied dataset is a test dataset with missing values and unverified real-world representativeness.	Results should not be treated as verified global EV market statistics.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Public website and Tableau Public are suitable for a demonstration and modest review traffic.	For higher traffic, evaluate a managed web host/CDN and an appropriately licensed analytics hosting option.
2	Data Storage	Current solution uses a CSV file connected to Tableau; no production database is configured.	Move validated, regularly refreshed data to a managed relational or cloud data warehouse if data volume and refresh needs grow.
3	Performance	Dashboard performance depends on workbook design, data volume, browser and Tableau hosting.	Optimise extracts, calculations, filters and chart complexity; test load times with larger representative data.
4	Security	Public-facing pages and Tableau views are intended for public viewing; no private user data workflow is implemented.	Keep sensitive data out of public files; use least-privilege access and secure authentication if private data or user accounts are added.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve data validation, document field definitions and refresh the CSV when a verified new dataset is available.	1–2 months after approval	More reliable and traceable analysis.
Phase 2	Connect a maintained database or warehouse and automate scheduled data refresh.	2–4 months after Phase 1	Less manual updating and support for larger datasets.

Future Roadmap – Continued

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 3	Improve dashboard performance, mobile layout and accessibility; run load and usability tests.	4–6 months after Phase 2	Faster, easier exploration across devices.
Phase 4	Add additional validated datasets, trend comparisons, regional analysis and documented stakeholder feedback.	6–9 months after Phase 3	Broader analysis and more useful decision support.

Implementation note: All roadmap items above are proposed enhancements. The current delivered solution remains a CSV-based Tableau Public dashboard, two Tableau stories and a static GitHub Pages website. No database, automated ingestion, authentication system or production-scale load testing is claimed.